DTC P0A12:00 [i-ELOOP]

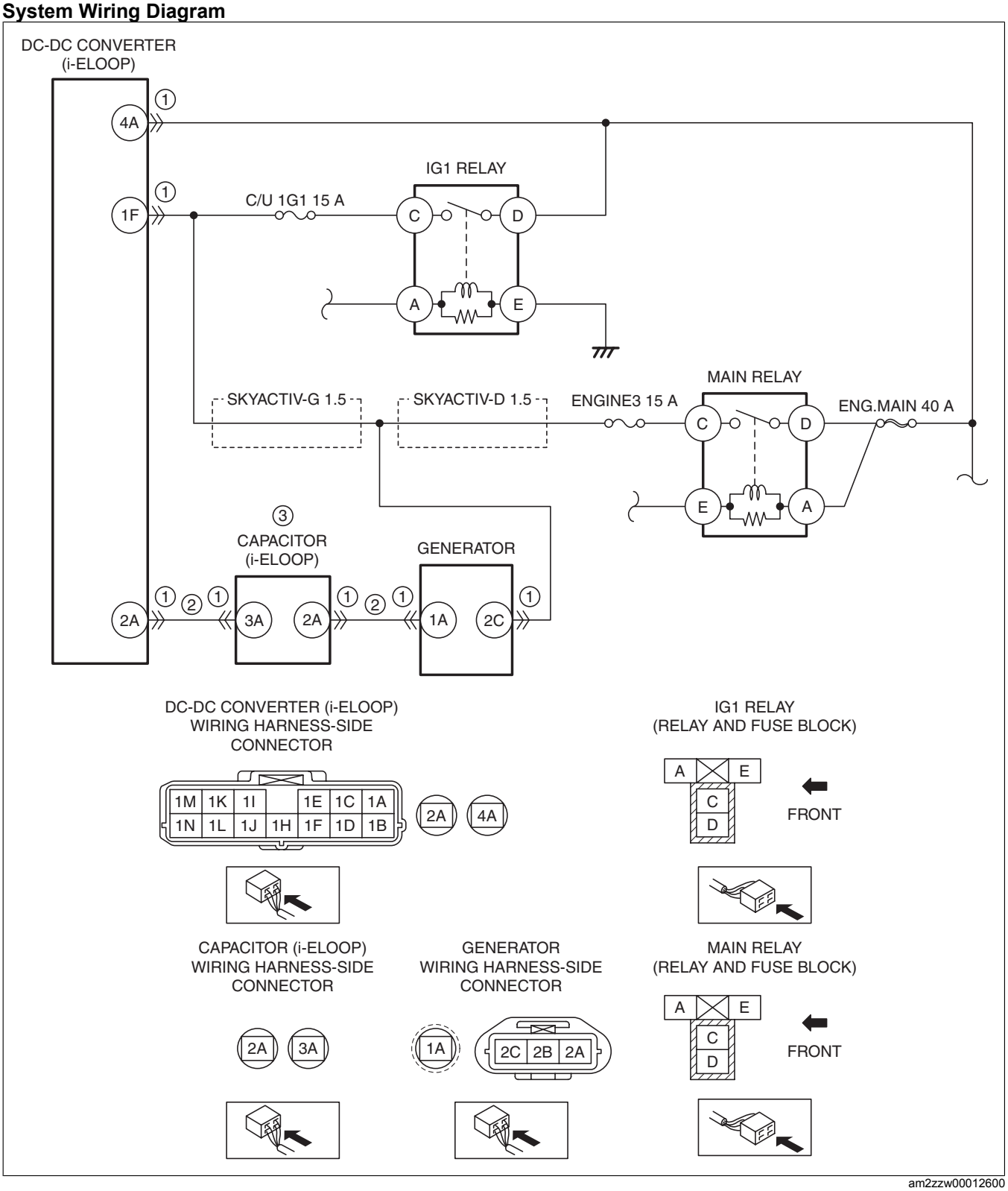
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Note
- To determine the malfunctioning part, proceed with the diagnostics from "Function Inspection Using M-MDS".
- If the capacitor (i-ELOOP) voltage is low after replacing the DC-DC converter (i-ELOOP), the i-ELOOP indicator light (green) flashes until the capacitor (i-ELOOP) voltage is **approx. 9.5 V**.

Details On DTCs

DESCRIPTION	DC-DC converter (i-ELOOP): low input	
DETECTION CONDITION	Determination conditions	• DC-DC converter (i-ELOOP) terminal 2A voltage is specified value or less.
	Preconditions	• Ignition switched ON
	Malfunction determination period	• **5 s** period
	Drive cycle	• 1
	Self test type	• CMDTC self test
	Sensor used	• DC-DC converter (i-ELOOP)
FAIL-SAFE FUNCTION	• Inhibits engine-stop by operating the i-stop function. (With i-stop)	
VEHICLE STATUS WHEN DTCs ARE OUTPUT	• Flashes i-stop warning light (amber). (With i-stop) • A warning message is displayed on the center display. (With center display) • Illuminates charging system warning light. • If the vehicle continues to be driven while the DTC is detected the battery will be depleted. — A malfunction occurs with an electrical device and the vehicle stops.	
POSSIBLE CAUSE	• Connector or terminal malfunction of the following parts: — PCM (PCM DTC may be detected simultaneously) — Generator — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) — Electrical load part which is connected to DC-DC converter (i-ELOOP) terminal 4A • Short to ground in wiring harness between the following terminals: — Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A — Capacitor (i-ELOOP) terminal 3A—DC-DC converter (i-ELOOP) terminal 2A — IG1 relay terminal C—Generator terminal 2C (PCM DTC P2503:00 may be detected simultaneously) (SKYACTIV-G 1.5) — Main relay terminal C—Generator terminal 2C (PCM DTC P2503:00 may be detected simultaneously) (SKYACTIV-D 1.5) • Open circuit in wiring harness between the following terminals: — IG1 relay terminal C—Generator terminal 2C (PCM DTC P2503:00 may be detected simultaneously) (SKYACTIV-G 1.5) — Main relay terminal C—Generator terminal 2C (PCM DTC P2503:00 may be detected simultaneously) (SKYACTIV-D 1.5) • DC-DC converter (i-ELOOP) malfunction	

System Wiring Diagram



am2zzw00012600

Function Explanation (DTC Detection Outline)

- With this diagnosis, a voltage of nearly **0 V** at the DC-DC converter (i-ELOOP) terminal 2A is detected for verification of vehicle malfunctions/safety assurance.

Repeatability Verification Procedure

1. Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].)
2. Start the engine.
3. Leave for **180 s** while idling with no electrical load.
4. Leave for **30 s** while idling with high electrical load.

PID Item/Simulation Item Used In Diagnosis
- Not applicable

Function Inspection Using M-MDS

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY RELATED SERVICE INFORMATION AVAILABILITY** • Verify related Service Information availability. • Is any related Service Information available?	Yes	Perform repair or diagnosis according to the available Service Information. • If the vehicle is not repaired, go to the next step.
		No	Go to the next step.
2	**PURPOSE: VERIFY IF POWER SUPPLY IS AFFECTED BY DTC RELATED TO DC-DC CONVERTER (i-ELOOP)** • Switch the ignition off, then ON (engine off). • Perform the DC-DC converter (i-ELOOP) DTC inspection using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the DTC P1628:11 or U3000:49 also present?	Yes	Go to the applicable DTC inspection. (See DTC P1628:11 [i-ELOOP].) (See DTC U3000:49 [i-ELOOP].) Go to the next step.
		No	Go to the next step.
3	**PURPOSE: VERIFY IF POWER SUPPLY IS AFFECTED BY DTC RELATED TO PCM** • Perform the PCM DTC inspection using the M-MDS. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) • Is the DTC P0A94:00 or P2503:00 also present?	Yes	Go to the applicable DTC inspection. (See DTC P0A94:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC P0A94:00 [SKYACTIV-D 1.5].) (See DTC P2503:00 [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See DTC P2503:00 [SKYACTIV-D 1.5].) Go to the troubleshooting procedure to perform the procedure from Step 1.
		No	Go to the troubleshooting procedure to perform the procedure from Step 1.

Troubleshooting Diagnostic Procedure
Intention of troubleshooting procedure
- Step 1—3
  - — Verifying malfunction related to DC-DC converter (i-ELOOP) 2A terminal
- Step 4—5
  - — Verifying/replacing malfunctioning part
- Step 6—7
  - — Verify that primary malfunction is resolved and there are no other malfunctions.

STEP	INSPECTION	RESULTS	ACTION
1	**PURPOSE: VERIFY IF CONNECTOR DAMAGE OF EACH PART AFFECTS DIAGNOSTIC RESULTS** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Remove the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Disconnect the connector of the following parts. — Generator — Capacitor (i-ELOOP) — DC-DC converter (i-ELOOP) • Inspect for poor connection (such as damaged/pulled-out pins, corrosion). • Is there any malfunction?	Yes	Repair or replace the connector and/or terminals, then go to Step 3.
		No	Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
2	**PURPOSE: VERIFY IF SHORT TO GROUND IN EACH WIRING HARNESS AFFECTS DIAGNOSTIC RESULTS** • Verify that the generator, capacitor (i-ELOOP), and DC-DC converter (i-ELOOP) connectors are disconnected. • Inspect for continuity between the following terminals (wiring harness-side) and body ground: — Capacitor (i-ELOOP) terminal 2A — Capacitor (i-ELOOP) terminal 3A • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between the following terminals: • Capacitor (i-ELOOP) terminal 2A—Generator terminal 1A • Capacitor (i-ELOOP) terminal 3A—DC-DC converter (i-ELOOP) terminal 2A **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Go to Step 4.
3	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].) • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	**Note** • A fuse is blown in the capacitor (i-ELOOP) due to excess current can be considered. Replace the capacitor (i-ELOOP). (See CAPACITOR (i-ELOOP) REMOVAL/INSTALLATION [i-ELOOP].) Go to Step 6.
		No	Go to Step 7.
4	**PURPOSE: VERIFY IF MALFUNCTION OCCURRING IS CAUSED BY WIRING HARNESS/CONNECTOR LOSS** • Have all the wiring harnesses/connectors which can be considered the cause of DTC P0A12:00 been inspected?	Yes	Go to the next step.
		No	Inspect all the related wiring harnesses/connectors. • If there is any malfunction: — Repair or replace the malfunctioning part according to the inspection results, then go to Step 6. • If there is no malfunction: — Go to the next step.
5	**PURPOSE: VERIFY DC-DC CONVERTER (i-ELOOP) REPLACEMENT RECORD** • Is the DC-DC converter (i-ELOOP) being replaced within the troubleshooting diagnostic procedure this time?	Yes	Go to the next step.
		No	Replace the DC-DC converter (i-ELOOP). (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-G 1.3, SKYACTIV-G 1.5)].) (See DC-DC CONVERTER (i-ELOOP) REMOVAL/INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) Go to the next step.

STEP	INSPECTION	RESULTS	ACTION
6	**PURPOSE: VERIFICATION OF VEHICLE REPAIR COMPLETION** • Always reconnect all disconnected connectors. • Reinstall the service plug. (See SERVICE PLUG DISCONNECTION/ CONNECTION [i-ELOOP].) • Reconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Start the engine and leave for **100 s or more**. • Clear the DTC for the DC-DC converter (i-ELOOP) using the M-MDS. (See CLEARING DTC [i-ELOOP].) • Implement the repeatability verification procedure. (See Repeatability Verification Procedure.) • Retrieve the DC-DC converter (i-ELOOP) DTCs using the M-MDS. (See DTC INSPECTION [i-ELOOP].) • Is the same DTC present?	Yes	Repeat the inspection from Step 4.
		No	Go to the next step.
7	**PURPOSE: VERIFY IF THERE IS ANY OTHER MALFUNCTION** • Is any other DTC or pending code stored?	Yes	Go to the applicable DTC inspection. (See DTC TABLE [i-ELOOP].)
		No	DTC troubleshooting completed.